

Drive for better vision



WE2 SRAM & Flash Usage

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Memory(SRAM: 1924KB) Usage

- WDMA1

- ❖ For CDM
- ❖ Default: 0 KB (Disable)

- WDMA2

- ❖ JPEG, 4X or 10X Compress
 - YUV Image Size/(4 or 10)
- ❖ Prerolling Buffer
 - As more as possible, for storing more preroll JPEG images.
- ❖ Default(HM0360, VGA, Mono):
 - $640 \times 480 \times 1 / 10 = 32\text{KB}$ (no Preroll Buffer)

- WDMA3

- ❖ YUV Image
- ❖ Width x Height x (1:YUV400, 1.5:YUV420)
- ❖ For ComputerVision or Image Process.
- ❖ Default(HM0360, VGA, Mono):
 - $640 \times 480 \times 1 = 300\text{KB}$

- TensorArena

- ❖ Max: 1924KB – WDMA1 – WDMA2 – WDMA3.
- ❖ Specify to .bss.tensor_arena
- ❖ e.g.

```
uint8_t tensor_arena[tensor_arena_size]__attribute__((section(".bss.tensor_arena"), aligned(16)));
```

Flash Layout

- Flash usage: 4K alignment
- Map
 - ❖ Memory Descriptor – 0x00000000 ~ 0x00000FFF
 - 4KB
 - ❖ 2nd Bootloader – 0x00001000 ~ 0x00010FFF
 - 64KB
 - ❖ Bootloader – 0x00011000 ~ 0x00024FFF
 - 80KB
 - ❖ Application – 0x00025000 ~ 0x0017FFFF
 - Including both CM55M_APP and M55S_APP.
 - ❖ TFLite Model – 0x00180000 ~ 0x00800000 (8MB Flash)
 - Corresponding XIP address: 0x3A180000, the address is defined in linker script.